Problem Formulation

Define δ_i as:
If X ∈ ***class I*** *and misclassified as* ***Class II*** then d(X) will be **<0** ⇒ δ_i = -1 ⇒ $d(X)\delta_i$ >0

If X ∈ ***class II*** *and misclassified as* ***Class I*** then d(X) will be **>0** ⇒ δ_i = +1 ⇒ $d(X)\delta_i$ >0

Redefine Error function as:

$$\text{Error} = \sum_{\text{all misclassified vectors (i)}} \delta_i W^T X_i$$

(W^TX_i) takes positive and negative values and δ_i has the same sign

Summation is always ***positive***

Minimize the Error using ***Gradient Descent*** *with step* ***η***

$$W^{\text{updated}} = W^{\text{old}} - \eta \frac{\partial \text{Error}}{\partial W} \quad \textbf{where} \quad \frac{\partial \text{Error}}{\partial W} = \sum_{\text{misclassified vectors}} \delta_i X_i$$

$$W^{\text{updated}} = W^{\text{old}} - \eta \sum_{\text{misclassified vectors ONLY (i)}} \delta_i X_i$$

Update after each sample

$$W^{\text{updated}} = W^{\text{old}} - \eta \delta_i X_i$$